

TRENT HAINES

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Education

University of Texas at Dallas

Master of Science in Computer Science

Graduated May 2023

Richardson, Texas

University of Texas at Dallas

Bachelor of Science in Computer Science

Graduated May 2022

Richardson, Texas

- 3.93 GPA
- Computing Scholar's Honors
- Full Ride Scholarship

Relevant Coursework

- Algorithm Analysis
- Distributed Computing
- C++ in UNIX Environment
- Statistics for AI/ML
- Database Systems
- Operating Systems

Experience

Roblox

Feb 2024 – Present

Software Engineer - Backend, Payment Systems

San Mateo, California

- Traced the full service communication network for Payment Search, revealing redundant paths and fragmented data sources. Rearchitected Payment Search to directly communicate with a new Elasticsearch-backed API, eliminating middlemen and establishing a single source of truth
- Redesigned the internal representation of "product", replacing an inflexible multi-table schema with a streamlined entities service. Led complex migration across upstream services, eliminating 6 SQL tables and 3 micro-services. Created a reusable Dual-Reading library during the project that became the company standard for data migrations.

Amazon

May 2022 – August 2022

Software Engineering Intern - Fullstack, Route Planning Optimization

Austin, Texas

- Developed an interactive web app enabling Amazon science teams to analyze delivery route execution patterns, guiding route-planning optimization decisions
- Analyzed route data requirements, identified 14 operational metrics, wrote database queries for extraction, developed backend APIs for data retrieval, and implemented frontend integration with interactive visualization for side by side route comparison

FireEye

May 2021 – August 2021

Software Engineering Intern - Backend, Telemetry

Dallas, Texas

- Improved the threat detection engine's reliability by migrating container infrastructure from ECS to EKS and introducing monitoring through Grafana dashboards and Prometheus metrics to track container performance, throttling events, and resource utilization

University of Texas at Dallas

Jun 2016 – Aug 2017

NanoTechnology Summer Research Intern (2x)

Richardson, Texas

- Developed applications and research on nylon artificial muscles and organic aerogels
- Work included in a successful NASA STTR Phase II grant application

Projects

Convex Hull Algorithm Visualizer | JavaScript, HTML, CSS

April 2022

- Created a react application that visualizes three different Convex Hull algorithms to help Computational Geometry students better understand and compare them. Students are able to place points down to create custom data-sets to better understand the strengths of the various algorithms in different scenarios

Blind Accessibility Chrome Extension | JavaScript, Python

April 2020

- Collaborated with a team to create a Chrome extension using Google's Cloud Vision API to provide text labeling for images on webpage. These labels can be read aloud using text-to-speech, or can assist in image captioning

Technical Skills

Languages: C++, C#, Java/Kotlin, Python, JavaScript/TSX

Frameworks and Libraries: Boost (C++), .NET/ASP.NET, gRPC/Protobuf

Tools: Git, Linux, Docker, Grafana